

Competitive Programming Notebook

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1 DP

1.1 Lis

```

1 // Find the length of the longest increasing
  subsequence in a given array
2 // Time complexity: O(n log n) where n is the length
  of the input array
3
4 int lis(vector<int>& arr) {
5     vector<int> dp;
6     for (int x : arr) {
7         auto it = lower_bound(all(dp), x);
8         if(it == dp.end()) dp.push_back(x);
9         else *it = x;
10    }
11    return dp.size();
12 }
```

1.2 Countingdp

```

1 // Counting DP (Sum of all ways / Unbounded Knapsack)
2 // Time Complexity: O(N * Target)
3 // Space Complexity: O(Target)
4 // Use case: Count number of ways to form a sum
5
6 int count_ways(vector<int>& items, int target) {
7
8     vector<int> dp(target + 1, 0);
9     dp[0] = 1;
10
11    for (int item : items) {
12        for (int i = item; i <= target; i++) {
13            dp[i] = (dp[i] + dp[i - item]) % MOD;
14        }
15    }
16
17    return dp[target];
18 }
```

1.3 Knapsack

```

1 // General take/not take dp problem
2 // Time Complexity: aprox. O(n * m)
3 // Space Complexity: O(n * m) / O(m)
4 // Use cases: maximize/minimize sum
5
6 // Top-down
7 int knapsack_top_down(vector<int>& v, vector<int>& w,
  int x){
8     int n = (int)v.size();
9     vector<vector<int>> dp(n, vector<int>(x+1, -1));
10    auto rec = [&](auto self, int i, int m) -> int {
11        if(i >= n || m == 0) return 0;
12        if(dp[i][m] != -1) return dp[i][m];
13        int take = 0, not_take = self(self, i+1, m);
14        if(m >= w[i]){
15            take = v[i] + self(self, i+1, m - w[i]);
16        }
17        return dp[i][m] = max(take, not_take);
18    };
19
20    return rec(rec, 0, x);
21 }
22
23 // Bottom-up
24 int knapsack_bottom_up(vector<int>& v, vector<int>& w,
  int x){
25     int n = (int)v.size();
26     vector<int> dp(x+1, 0);
27 }
```

```

28     for(int i=0; i < n; i++){
29         for(int m=x; m >= w[i]; m--){
30             dp[m] = max(dp[m], v[i] + dp[m - w[i]]);
31         }
32     }
33
34     return dp[x];
35 }
```

2 Math

2.1 Factorial

```

1 // Time Complexity: O(N) for precompute, O(1) for nCr
2 // Space Complexity: O(N)
3 // Use cases: calculating combinations, permutations,
  probabilities
4
5 const int MAXN, MOD;
6 vector<int> fact(MAXN), ifact(MAXN);
7
8 void precompute(){
9
10    fact[0] = 1; ifact[0] = 1;
11    for(int i=1; i<MAXN; i++){
12        fact[i] = (i * fact[i-1]) % MOD;
13    }
14
15    ifact[MAXN-1] = fexp(fact[MAXN-1], MOD-2, MOD);
16    for(int i=MAXN-2; i >= 1; i--){
17        ifact[i] = (ifact[i+1]*(i+1)) % MOD;
18    }
19 }
20
21 int choose(int n, int r){
22     if(r < 0 or r > n) return 0;
23     int num = fact[n];
24     int den = (ifact[r] * ifact[n - r]) % MOD;
25     return (num * den) % MOD;
26 }
```

2.2 Fastexponentiation

```

1 // Calculates (x^n) % MOD
2 // Time Complexity: O(log n)
3 // Space Complexity: O(1)
4
5 int fexp(int x, int b, int MOD){
6     int res = 1;
7
8     while(b > 0){
9         if(b & 1) res = (res * x) % MOD;
10        x = (x * x) % MOD;
11        b >>= 1;
12    }
13    return res;
14 }
```

2.3 Crivo

```

1 // Find all prime numbers up to a given limit using
  the Sieve of Eratosthenes algorithm
2 // Time complexity: O(n log log n) where n is the
  limit
3 // This idea can also be used to construct a table
  for the smallest prime factors, O(log N) per
  query
4
5 vector<int> findPrimes(int limit) {
6     vector<bool> isPrime(limit + 1, true);
7     isPrime[0] = isPrime[1] = false;
```

```

8
9 for (int i = 2; i * i <= limit; i++) {
10     if (isPrime[i]) {
11         for (int j = i * i; j <= limit; j += i) {
12             isPrime[j] = false;
13         }
14     }
15 }
16
17 vector<int> primes;
18 for (int i = 2; i <= limit; i++) {
19     if (isPrime[i]) {
20         primes.push_back(i);
21     }
22 }
23 return primes;
24 }
25
26 vector<int> buildSPF(int limit){
27     vector<int> spf(limit+1);
28     iota(spf.begin(), spf.end(), 0);
29     for(int i = 2; i * i <= limit; i++) {
30         if(spf[i] == i){
31             for(int j = i * i; j <= limit; j += i) {
32                 if(spf[j] == j) spf[j] = i;
33             }
34         }
35     }
36
37     return spf;
38 }
39
40 vector<int> getFactors(vector<int>& spf, int x){
41     vector<int> factors;
42     while(x > 1){
43         factors.push_back(spf[x]);
44         x /= spf[x];
45     }
46
47     return factors;
48 }

```

2.4 Matrixexponentiation

```

1 // Calculate the Nth Fibonacci number using matrix
  exponentiation.
2 // Time Complexity: O(log N)
3 // Space Complexity: O(1)
4
5 void matrixExponentiation(vector<vector<long long>>&
  matrix, int power) {
6     int n = matrix.size();
7     vector<vector<long long>> result(n, vector<long
  long>(n, 0));
8
9     // Initialize result as the identity matrix
10    for (int i = 0; i < n; i++) {
11        result[i][i] = 1;
12    }
13
14    while (power) {
15        if (power % 2 == 1) {
16            matrixMultiplication(result, matrix,
  result);
17        }
18        matrixMultiplication(matrix, matrix, matrix);
19        power /= 2;
20    }
21
22    matrix = result;
23 }

```

2.5 Matrixmultiplication

```

1 // Multiply two matrices A and B, and return the
  result in matrix C.
2 // Time Complexity: O(n^3)
3 // Space Complexity: O(n^2)
4
5 vector<vector<int>> matrixMultiplication(const vector
  <vector<int>>& A, const vector<vector<int>>& B) {
6     int n = A.size();
7     vector<vector<int>> C(n, vector<int>(n, 0)); //
  Initialize result matrix with zeros
8
9     for (int i = 0; i < n; i++) {
10        for (int j = 0; j < n; j++) {
11            for (int k = 0; k < n; k++) {
12                C[i][j] += A[i][k] * B[k][j];
13            }
14        }
15    }
16    return C;
17 }

```

2.6 Extendedeuclidean

```

1 // Extended Euclidean Algorithm
2 // Returns (gcd(a, b), x, y) such that a*x + b*y =
  gcd(a, b)
3
4 int extendedEuclidean(int a, int b, int &x, int &y){
5     if(a == 0){
6         x = 0;
7         y = 1;
8         return b;
9     }
10
11    int x1, y1;
12    int gcd = extendedEuclidean(b % a, a, x1, y1);
13    x = y1;
14    y = x1 - (b / a) * y1;
15    return gcd;
16 }

```

3 String

3.1 Hash

```

1 // String Hash template
2 // constructor(s) - O(|s|)
3 // query(l, r) - returns the hash of the range [l,r]
  from left to right - O(1)
4 // query_inv(l, r) from right to left - O(1)
5 // patrocinado por tiagodfs
6
7 mt19937 rng(time(nullptr));
8
9 struct Hash {
10     const int X = rng();
11     const int MOD = 1e9+7;
12     int n; string s;
13     vector<int> h, hi, p;
14     Hash() {}
15     Hash(string s): s(s), n(s.size()), h(n), hi(n), p
  (n) {
16         for (int i=0;i<n;i++) p[i] = (i ? X*p[i-1]:1)
  % MOD;
17         for (int i=0;i<n;i++)
18             h[i] = (s[i] + (i ? h[i-1]:0) * X) % MOD;
19         for (int i=n-1;i>=0;i--)
20             hi[i] = (s[i] + (i+1<n ? hi[i+1]:0) * X)
  % MOD;

```

```

21     }
22     int query(int l, int r) {
23         int hash = (h[r] - (l ? h[l-1]*p[r-l+1]%MOD :
24             0));
25         return hash < 0 ? hash + MOD : hash;
26     }
27     int query_inv(int l, int r) {
28         int hash = (hi[l] - (r+1 < n ? hi[r+1]*p[r-l
29             +1] % MOD : 0));
30         return hash < 0 ? hash + MOD : hash;
31     }
32 }

```

4 Graph

4.1 Kruskal

```

1 // Kruskal algorithm for finding the Minimum Spanning
  Tree
2 // Time Complexity: O(E log E)
3 // Space complexity: O(V + E)
4 // Use cases: connect nodes with min/max cost,
  clustering, minimax path
5
6 // edges: {weight, u, v}
7 vector<vector<pair<int, int>>> kruskalMST(vector<
  array<int, 3>>& edges, int n){
8     int m = (int)edges.size();
9     sort(edges.begin(), edges.end());
10    vector<vector<pair<int, int>>> mst(n+1);
11    DSU dsu(n+1); int cost = 0;
12    for(int i=0; i<m; i++){
13        int w = edges[i][0], a = edges[i][1], b =
14        edges[i][2];
15        if(dsu.merge(a, b)){
16            mst[a].push_back({w, b});
17            mst[b].push_back({w, a});
18            cost += w;
19        }
20    }
21    return mst;
22 }

```

4.2 Eulereanpath

```

1 // Find the Eulerian path in a graph using Hierholzer
  's algorithm
2 // Time complexity: O(E) where E is the number of
  edges in the graph
3 // Use cases: visit every edge once, DNA sequencing,
  puzzles, drawing figures
4
5 vector<int> findEulerianPath(vector<vector<int>>& adj
  , int start) {
6
7     vector<int> path;
8     stack<int> s;
9     s.push(start);
10
11     while (!s.empty()) {
12         int i = s.top();
13         if (adj[i].empty()) {
14             path.push_back(i);
15             s.pop();
16         } else {
17             int u = adj[i].back(); adj[i].pop_back();
18             s.push(u);
19         }
20     }
21
22     reverse(path.begin(), path.end());

```

```

23     return path;
24 }

```

4.3 Dijkstra

```

1 // Find shortest path from a source vertex to all
  other vertices in a graph with non-negative edge
  weights
2 // Time Complexity: O((V + E) log V)
3 // Space Complexity: O(V)
4 // Use cases: shortest path in non-negative weighted
  graphs
5
6 vector<int> dijkstra(vector<vector<pair<int, int>>>&
  adj, int src) {
7
8     int n = adj.size() - 1;
9     vector<int> dist(n+1, INF);
10    dist[src] = 0;
11
12    priority_queue<pair<int, int>, vector<pair<int,
13        int>>, greater<pair<int, int>>> pq; // Min-heap
14    priority queue
15    pq.push({0, src});
16
17    while (!pq.empty()) {
18
19        auto [d, i] = pq.top();
20        pq.pop();
21
22        if(d > dist[i]) continue;
23
24        for (auto [u, w] : adj[i]) {
25
26            if (dist[i] + w < dist[u]) {
27                dist[u] = dist[i] + w;
28                pq.push({dist[u], u});
29            }
30        }
31
32    }
33
34    return dist;
35 }

```

4.4 Bellmanford

```

1 // Find the shortest path from a source vertex to all
  other vertices in a graph with negative weight
  edges and detect negative weight cycles.
2 // Time Complexity: O(V * E)
3 // Space Complexity: O(V)
4 // Use cases: SSSP with negative edges, negative
  cycle detection, system of difference constraints
  , shortest path with at most K edges
5
6 bool bellmanFord(vector<vector<pair<int, int>>>& adj,
  int src) {
7
8     int n = adj.size() - 1;
9     vector<int> dist(n+1, INF);
10    dist[src] = 0;
11
12    // Relax all edges n-1 times
13    for (int i = 1; i < n; i++) {
14        for (int u = 1; u <= n; u++) {
15            for (auto& [v, w] : adj[u]) {
16                if (dist[u] != INF and dist[u] + w <
17                    dist[v]) {
18
19                        dist[v] = dist[u] + w;
20                    }
21            }
22        }
23    }
24
25    // Detect negative weight cycles
26    for (int u = 1; u <= n; u++) {
27        for (auto& [v, w] : adj[u]) {
28            if (dist[u] != INF and dist[u] + w <
29                dist[v]) {
30                return false;
31            }
32        }
33    }
34
35    return true;
36 }

```

```

21     }
22
23     // Check for negative weight cycles
24     for (int u = 1; u <= n; u++) {
25         for (auto& [v, w] : adj[u]) {
26             if (dist[u] != INF and dist[u] + w < dist
[v]) {
27                 // Graph contains negative weight
28                 cycle
29                 return true;
30             }
31         }
32     }
33     return false;
34 }

```

4.5 Floydwarshall

```

1 // Find the shortest paths in a weighted graph with
  positive or negative edge weights (but with no
  negative cycles).
2 // Time Complexity:  $O(V^3)$ 
3 // Space Complexity:  $O(V^2)$ 
4 // Use cases: reachability, minimax distance, global
  negative cycle detection (dist[i][i] < 0)
5
6 void floydWarshall(vector<vector<int>>& dist) { //
  dist(INF)
7
8   int n = dist.size() - 1;
9   for(int i = 1; i <= n; i++) dist[i][i] = 0;
10
11   for (int k = 1; k <= n; k++) {
12       for (int i = 1; i <= n; i++) {
13           for (int j = 1; j <= n; j++) {
14               if (dist[i][k] != INF and dist[k][j]
!= INF) {
15                   dist[i][j] = min(dist[i][j], dist
[i][k] + dist[k][j]);
16               }
17           }
18       }
19   }
20 }

```

4.6 Kahn

```

1 // Topological Sort (Kahn's Algorithm)
2 // Time Complexity:  $O(V + E)$ 
3 // Space Complexity:  $O(V)$ 
4 // Use cases: Dependency resolution, cycle detection
  in directed graphs, DP on DAGs
5
6 vector<int> TopologicalSort(vector<vector<int>>& adj,
  int n){
7     vector<int> degree(n+1, 0);
8     for(int i=1; i<=n; i++){
9         for(auto u: adj[i]) degree[u]++;
10    }
11
12    // priority_queue in case of lexicographically
  smallest answer
13    queue<int> q; vector<int> ans;
14    for(int i=1; i<=n; i++){
15        if(degree[i] == 0){
16            q.push(i); ans.push_back(i);
17        }
18    }
19
20    while(!q.empty()){
21

```

```

22         int i = q.front(); q.pop();
23         for(auto u: adj[i]){
24             degree[u]--;
25             if(degree[u] == 0){
26                 q.push(u); ans.push_back(u);
27             }
28         }
29     }
30
31     if((int)ans.size() != n) return {};
32     return ans;
33 }
34 }

```

5 Geometry

5.1 Point

```

1
2 const double EPS = 1e-9;
3 // EPS = 0 -> int
4
5 struct Point {
6     double x, y;
7
8     Point() : x(0), y(0) {}
9     Point(double x, double y) : x(x), y(y) {}
10
11     Point operator+(const Point& p) const { return
Point(x + p.x, y + p.y); }
12     Point operator-(const Point& p) const { return
Point(x - p.x, y - p.y); }
13     Point operator*(double c) const { return Point(x
* c, y * c); }
14     Point operator/(double c) const { return Point(x
/ c, y / c); }
15
16     bool operator<(const Point& p) const {
17         if (abs(x - p.x) > EPS) return x < p.x;
18         return y < p.y - EPS;
19     }
20
21     bool operator==(const Point& p) const {
22         return abs(x - p.x) <= EPS && abs(y - p.y) <=
EPS;
23     }
24 };
25
26 // > 0: B esq. vetor OA, < 0: B dir. vetor OA, 0:
  Colineares
27 double cross_product(Point O, Point A, Point B) {
28     return (A.x - O.x) * (B.y - O.y) - (A.y - O.y) *
(B.x - O.x);
29 }
30
31 // distancia euclidiana
32 double dist(Point A, Point B) {
33     return hypot(A.x - B.x, A.y - B.y);
34 }
35
36 // distancia quadrada
37 double dist2(Point A, Point B) {
38     return (A.x - B.x) * (A.x - B.x) + (A.y - B.y) *
(A.y - B.y);
39 }
40
41 // produto escalar, dot == 0 perpendiculares
42 double dot_product(Point O, Point A, Point B) {
43     return (A.x - O.x) * (B.x - O.x) + (A.y - O.y) *
(B.y - O.y);
44 }
45

```

```

46 int get_half(Point O, Point P) {
47     if (P.y - O.y > EPS or (abs(P.y - O.y) <= EPS and
48         P.x - O.x > EPS)) return 1;
49     return 0;
50 }
51 // ordenacao por perimetro
52 void sort_by_angle(vector<Point>& p, Point center) {
53     sort(p.begin(), p.end(), [&](const Point& A,
54         const Point& B) {
55         int halfA = get_half(center, A);
56         int halfB = get_half(center, B);
57
58         if (halfA != halfB) {
59             return halfA > halfB;
60         }
61
62         double cross = cross_product(center, A, B);
63         if (abs(cross) > EPS) {
64             return cross > 0;
65         }
66
67         return dist2(center, A) < dist2(center, B) -
68             EPS;
69     });
70 // shoelace formula, vÃrtices do polÃgono ordenados
71 // pelo perÃmetro.
72 double polygon_area(const vector<Point>& p) {
73     double area = 0.0;
74     int n = p.size();
75
76     for (int i = 0; i < n; i++) {
77         int next_i = (i + 1) % n;
78         area += (p[i].x * p[next_i].y) - (p[next_i].x
79             * p[i].y);
80     }
81
82     // return abs(area) -> int
83     return abs(area) / 2.0;
84 }
85 // aux func
86 int orientation(Point O, Point A, Point B) {
87     double val = cross_product(O, A, B);
88     if (abs(val) <= EPS) return 0; // Colinear
89     return (val > 0) ? 1 : -1; // 1: Esquerda,
90     -1: Direita
91 }
92 // 2. P in seg AB?
93 bool on_segment(Point P, Point A, Point B) {
94     return P.x >= min(A.x, B.x) - EPS && P.x <= max(A
95         .x, B.x) + EPS &&
96         P.y >= min(A.y, B.y) - EPS && P.y <= max(A
97         .y, B.y) + EPS;
98 }
99 // Ab seg cross CD?
100 bool segments_intersect(Point A, Point B, Point C,
101     Point D) {
102     // Calcula as 4 orientaÃÃes necessÃrias
103     int o1 = orientation(A, B, C);
104     int o2 = orientation(A, B, D);
105     int o3 = orientation(C, D, A);
106     int o4 = orientation(C, D, B);
107
108     if (o1 != o2 && o3 != o4) {
109         return true;
110     }
111
112     if (o1 == 0 && on_segment(C, A, B)) return true;

```

```

// C toca AB
if (o2 == 0 && on_segment(D, A, B)) return true;
// D toca AB
if (o3 == 0 && on_segment(A, C, D)) return true;
// A toca CD
if (o4 == 0 && on_segment(B, C, D)) return true;
// B toca CD

return false; // NÃo se cruzam

```

6 .template

6.1 Template

```

1 #include<bits/stdc++.h>
2
3 using namespace std;
4
5 #define int long long
6 #define vi vector<int>
7 #define pii pair<int, int>
8 #define all(x) (x).begin(), (x).end()
9 #define debug(x...) cout<<"["#x"]": ",[(auto...$){{{
10     cout<<$<<" "; }},...]);}(x),cout<<endl
11
12 void solve(){
13 }
14
15 signed main(){
16     ios::sync_with_stdio(false);
17     cin.tie(NULL);
18     int t = 1;
19     //cin >> t;
20     while(t--){ solve();
21     }
22 }

```

7 DS

7.1 Psum2d

```

1 // Psum2D
2 // Time Complexity: Query O(1)
3 // Space Complexity: O(N*M)
4 // Use cases: static 2d range sum, fixed-size window
5 // search, max sum submatrix
6
7 int n, m;
8 vector<vector<int>> arr(n+1, vector<int>(m+1, 0));
9 vector<vector<int>> psum(n+1, vector<int>(m+1, 0));
10
11 for (int i = 1; i <= n; i++) {
12     for (int j = 1; j <= m; j++) {
13         cin >> arr[i][j];
14         psum[i][j] = arr[i][j]
15             + psum[i][j-1]
16             + psum[i-1][j]
17             - psum[i-1][j-1];
18     }
19 }
20
21 // canto superior esquerdo (x1, y1) canto inferior
22 // direito (x2, y2)
23 int x1, y1, x2, y2;
24 int res = psum[x2][y2]
25     - psum[x1 - 1][y2]
26     - psum[x2][y1 - 1]
27     + psum[x1 - 1][y1 - 1];

```

7.2 Sparsetable

```

1 // Sparse Table
2 // Time Complexity: O(n * log n) build, O(1) query
3 // Space complexity: O(n * log n)
4 // Idempotent operation f(a, a) = a (min, max, gcd,
5 // Use cases: idempotent functions range queries
6
7 struct SparseTable{
8
9     int n, p;
10    vector<vector<int>> st;
11
12    SparseTable(vector<int> &arr){
13        n = (int) arr.size();
14        p = __lg(n);
15        st.assign(p+1, vector<int>(n, 0));
16
17        for(int j=0; j<n; j++) st[0][j] = arr[j];
18        for(int i = 1; i <= p; i++){
19            int k = (1ll << (i-1));
20            int lim = n - 2*k + 1;
21            for(int j=0; j<lim; j++){
22                st[i][j] = op(st[i-1][j], st[i-1][j+k]);
23            }
24        }
25    }
26
27    int query(int l, int r){
28        int len = r - l + 1;
29        int p_local = __lg(len), k = (1ll << p_local);
30        return op(st[p_local][l], st[p_local][r - k + 1]);
31    }
32 }
33 };

```

7.3 Segtree

```

1 template <typename T>
2
3 struct SegTree{
4     int n;
5     vector<T> tree;
6     const T NEUTRO = 1e18;
7
8     T merge(T a, T b) {return min(a, b);}
9     SegTree(int n){
10        this->n = n;
11        tree.assign(2*n, NEUTRO);
12    }
13
14    void build(vector<T>& v){
15        for(int i=0; i<n; i++) tree[n+i] = v[i];
16        for(int i=n-1; i>0; i--) tree[i] = merge(tree[2*i], tree[2*i + 1]);
17    }
18
19    void update(int pos, T val){
20        pos += n;
21        tree[pos] = val;
22        for(pos /= 2; pos > 0; pos /= 2) tree[pos] = merge(tree[pos*2], tree[pos*2 + 1]);
23    }
24
25    T query(int l, int r){
26        T res_l = NEUTRO, res_r = NEUTRO;
27        for(l += n, r += n; l <= r; l /= 2, r /= 2){
28            if(l%2 != 0) res_l = merge(res_l, tree[l++]);

```

```

29            if(r%2 == 0) res_r = merge(tree[r--],
30            res_r);
31        }
32        return merge(res_l, res_r);
33    }
34 };

```

7.4 Bit2d

```

1 // Binary Indexed Tree(BIT) 2D
2 // Time complexity: query and update log N * log M
3 // Space complexity: O(N*M)
4 // Use cases: dynamic 2D prefix sums, counting active
5 // points in a dynamic rectangle
6
7 struct BIT2D {
8
9     int n, m; vector<vector<int>> bit;
10    BIT2D(int n, int m) : n(n), m(m) {
11        bit.assign(n+1, vector<int>(m+1, 0));
12    }
13
14    void update(int x, int y, int val) {
15        for (int i = x; i <= n; i += i & -i) {
16            for (int j = y; j <= m; j += j & -j) {
17                bit[i][j] += val;
18            }
19        }
20    }
21
22    int query(int x, int y) {
23        int sum = 0;
24        for (int i = x; i > 0; i -= i & -i) {
25            for (int j = y; j > 0; j -= j & -j) {
26                sum += bit[i][j];
27            }
28        }
29        return sum;
30    }
31
32    int queryRange(int x1, int y1, int x2, int y2) {
33        return query(x2, y2)
34            - query(x1 - 1, y2)
35            - query(x2, y1 - 1)
36            + query(x1 - 1, y1 - 1);
37    }
38 };

```

7.5 Dsu

```

1 // Disjoint Set Union(DSU)
2 // Time Complexity: aprox. O(1)
3 // Space Complexity: O(n)
4 // Use cases: connected components, cycle detection,
5 // offline edge deletion
6
7 struct DSU{
8
9     vector<int> parent, size;
10
11    DSU(int n){
12        parent.assign(n+1, 0);
13        size.assign(n+1, 1);
14        iota(parent.begin(), parent.end(), 0);
15    }
16
17    int find(int v){
18        if(v == parent[v]) return v;
19        return parent[v] = find(parent[v]);
20    }

```

```

20
21 bool same(int a, int b){
22     return find(a) == find(b);
23 }
24
25 bool merge(int a, int b){
26     a = find(a);
27     b = find(b);
28     if(a != b){
29         if(size[a] < size[b]) swap(a, b);
30         parent[b] = a;
31         size[a] += size[b];
32         return true;
33     }
34     return false;
35 }
36
37 };

```

7.6 Bit

```

1 // Binary Indexed Tree(BIT)
2 // Time complexity: query and update log N
3 // Space complexity: O(N)
4 // Use cases: dynamic psum, inversion counting,
5 // dynamic multiset
6
7 struct BIT{
8     int n; vector<int> bit;
9     BIT(int n) : n(n){
10         bit.assign(n+1, 0);
11     }
12
13     void update(int i, int x){
14         while(i <= n){
15             bit[i] += x;
16             i += i & -i;
17         }
18     }
19
20     int query(int i){
21         int sum = 0;
22         while(i > 0){
23             sum += bit[i];
24             i -= i & -i;
25         }
26         return sum;
27     }
28
29     int queryRange(int l, int r){
30         return query(r) - query(l-1);
31     }
32 }

```

```

32
33 // retorna o índice do k-ésimo elemento da bit
34 int find_kth(int k){
35
36     int idx = 0;
37     int max_bit = __lg(n);
38
39     for(int i = max_bit; i >= 0; i--){
40         int next_idx = idx + (1ll << i);
41         if(next_idx <= n and bit[next_idx] < k){
42             idx = next_idx;
43             k -= bit[next_idx];
44         }
45     }
46
47     return idx+1;
48 }
49
50 };

```

7.7 Monotonicstack

```

1 // Monotonic Stack, next greater number in a circular
2 // array
3 // Time Complexity: O(N)
4 // Space Complexity: O(N)
5 // Use cases: ring problems, finding next larger
6 // obstacle in a loop
7
8 for (int i = 0; i < 2 * n; i++) {
9     int num = a[i % n];
10    while (!st.empty() and a[st.top()] < num) { //
11        remove os menores
12        res[st.top()] = num; st.pop();
13    }
14
15    if (i < n) st.push(i);
16 }
17
18 // Monotonic Stack, next smaller element in a linear
19 // array
20 // Use cases: largest rectangle in histogram,
21 // subarray contribution
22
23 for (int i = 0; i < n; i++) {
24     while (!st.empty() and a[st.top()] > a[i]) { //
25         remove os maiores
26         res[st.top()] = a[i];
27         st.pop();
28     }
29     st.push(i);
30 }
31 }

```